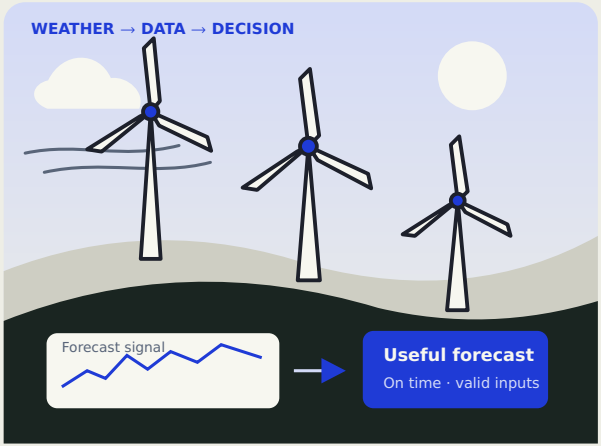
LECTURE 01 / THEORY

Introduction to MLOps

A good model is a beginning.
A reliable system is the goal.

MODELS / LIFECYCLE / RESPONSIBILITY



Three layers of the same problem.

01
Model to system

What must work around the predictor?

02
Operating loop

How does a change become a release?

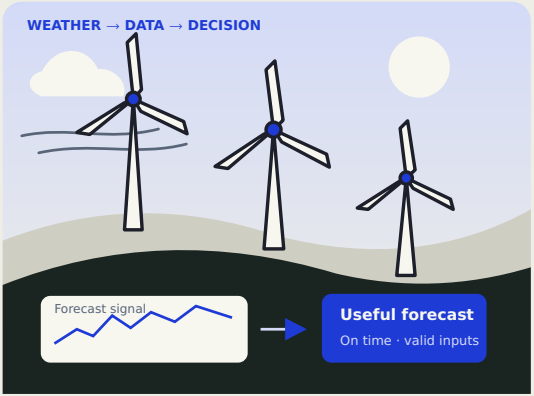
03
Responsibility

Who observes, decides and recovers?

Understand the relationships before choosing the tools.

Lecture structure; practical activities are separate.

The forecast must exist when it is needed.



ILLUSTRATIVE WIND-ENERGY CASE

A recurring decision, not a saved notebook.

HORIZON	Next-hour generation
CADENCE	Every 15 minutes
OUTPUT	Energy in MWh

S1 slides 16–17; horizon, cadence and artwork are illustrative.

A model is only one component.

ML SYSTEM

Data + pipelines
Infrastructure
Monitoring + people

APPLICATION

An output someone uses in a decision.

MODEL

Features
→ prediction

Nested responsibilities, not measured proportions of code or cost.

S1 slides 14–15; S2 pages 16–17; Sculley et al., 2015.

THE DEFINITION

MLOps connects development and operation.

Practices for building, releasing and maintaining ML systems reliably and efficiently.

Repeatable

Observable

Controlled

S1 slide 15; S2 page 37; Google Cloud. Working paraphrase.

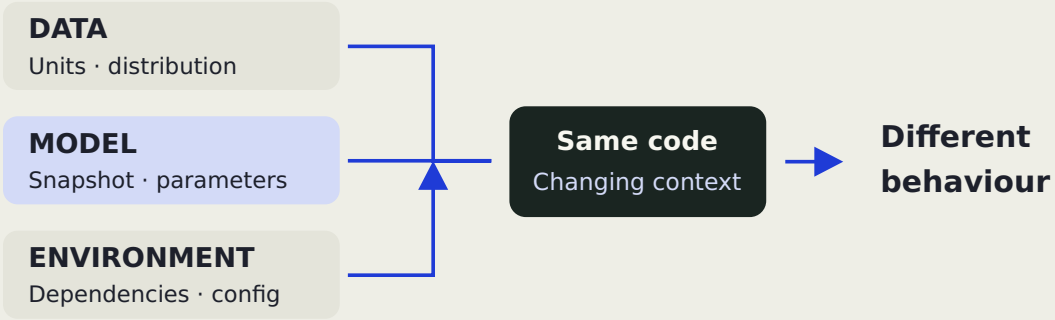
Accuracy is only one dimension.



An accurate forecast delivered too late is not a useful service.

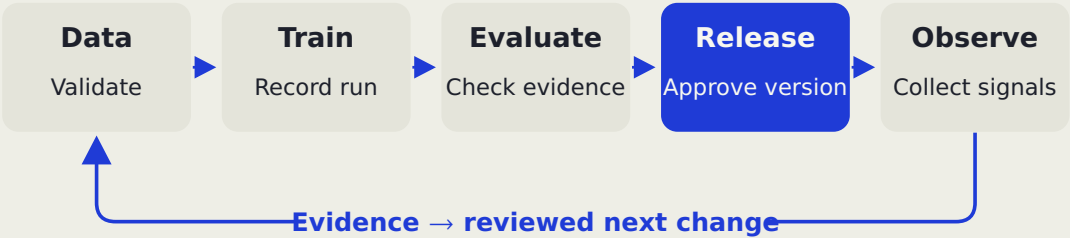
S1 slides 16–17, 54, 57–58; S2 pages 27, 83–84.

Fixed code does not mean fixed behaviour.



Test software contracts, data meaning and model behaviour.

Deployment is a milestone, not the finish.



Feedback informs a decision; it is not an automatic retraining command.

S1 slides 19–28; S2 pages 19–28. Simplified lifecycle.

Begin with the decision, not the algorithm.

THE USER CONTEXT

What changes because of this forecast?

USE	Operator plans generation
OUTPUT	Next-hour energy in MWh
TIMING	Before the decision
FALLBACK	Agreed response to missing inputs

S1 slides 52–58; S2 pages 76, 82–84. Illustrative decision context.

A valid number can be the wrong input.

EXPECTED UNITS

10

m/s

=

Same wind.
Different units.

NEW PROVIDER FEED

36

km/h

The numeric type is unchanged. The units contract is not.

Original theoretical example; 10 m/s = 36 km/h.

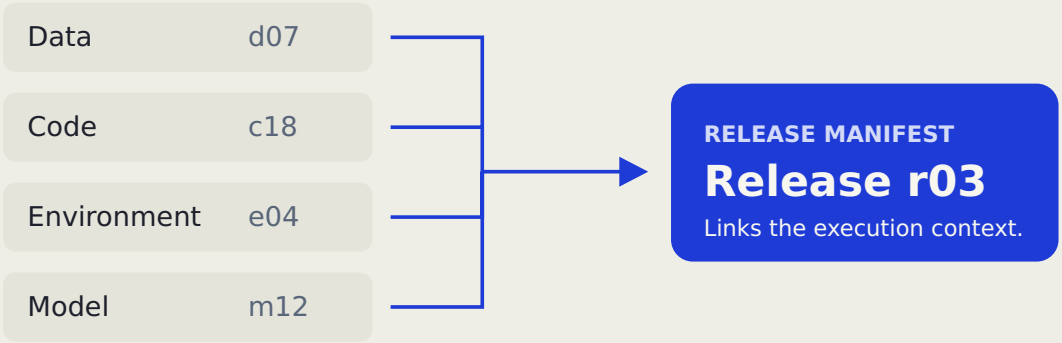
A result needs an execution context.



A metric without provenance cannot explain what changed.

S1 experiment-platform concepts, especially slide 63; identifiers are illustrative.

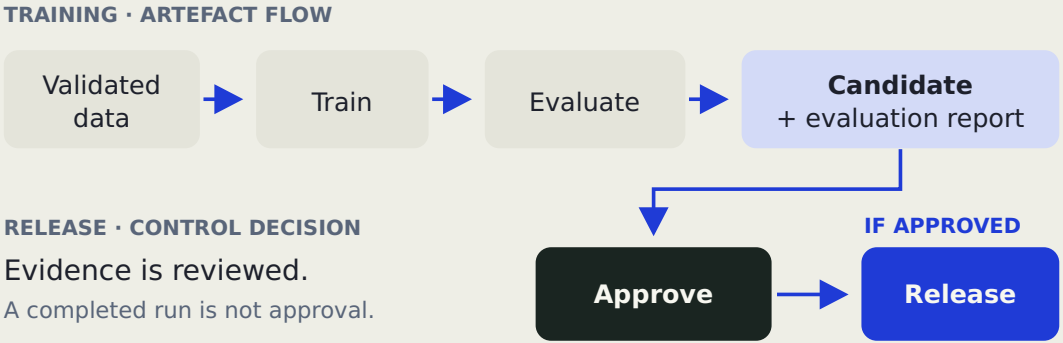
Version more than the source code.



Traceability enables investigation; it does not prove bitwise determinism.

S1 platform and reproducibility themes; illustrative identifiers.

Training produces a candidate, not approval.



Artefact flow and release control are different responsibilities.

S1 pipeline and deployment concepts; S2 pages 22–28.

Deploying changes the version. Serving uses it.

DEPLOYMENT

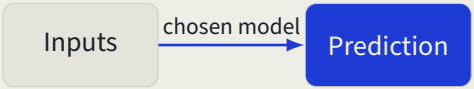
Select the running release.



A change in the environment.

SERVING

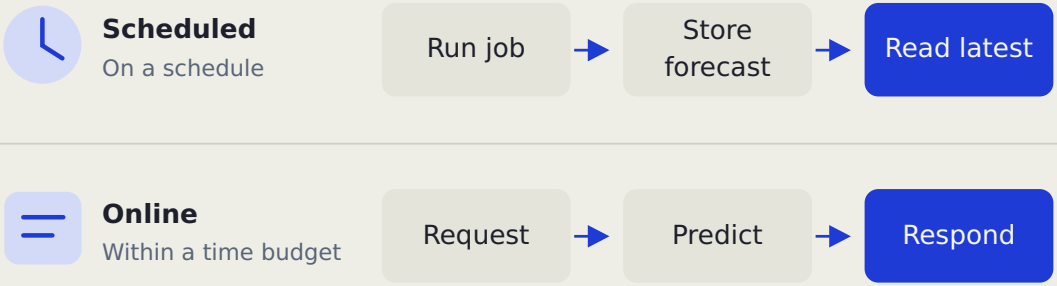
Execute a prediction.



A recurring execution path.

S1 and S2 serving/deployment material; S2 pages 21–22.

The decision timing shapes the serving pattern.



A fast response can still contain a stale forecast.

S1 serving concepts; S2 page 21. Conceptual comparison.

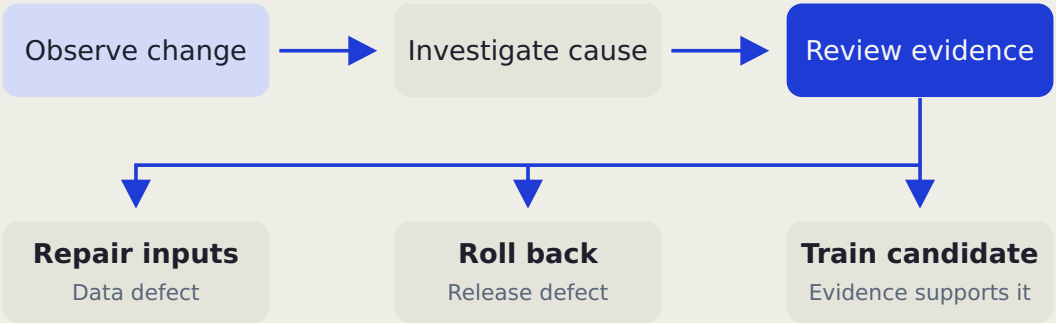
A healthy endpoint can hide a failing model.

Inputs	Meaning + freshness
Service	Errors + latency + availability
Model	Behaviour; error when outcomes arrive
Outcome	Usefulness for the real decision

Model error needs actual outcomes; those outcomes may arrive later.

S1 slides 27, 54; S2 pages 27, 76–84.

A drift alert is not a retraining command.



A distribution change does not by itself prove degraded performance.

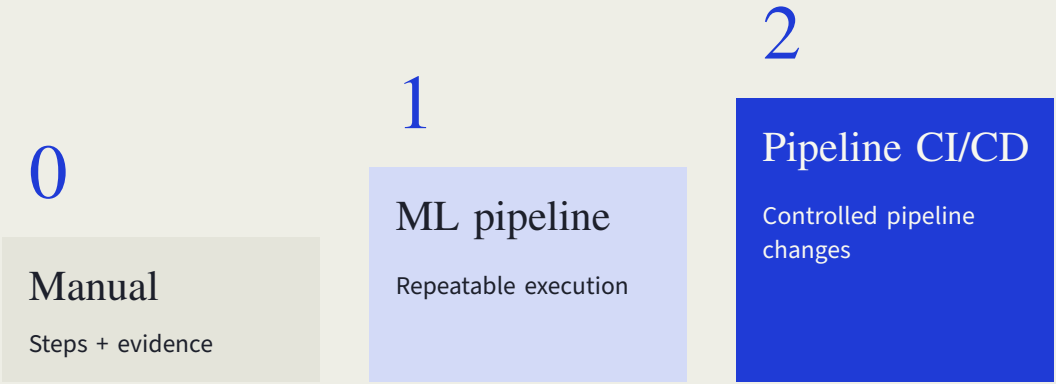
Conceptual synthesis of S1/S2 monitoring and pipeline material.

CI, CD and CT control different changes.

CI	Code change	→	Tests + contract checks
CD	Release change	→	Validated delivery + approval
CT	Training trigger	→	Repeatable candidate training

Here CD means continuous delivery; promotion can require approval.

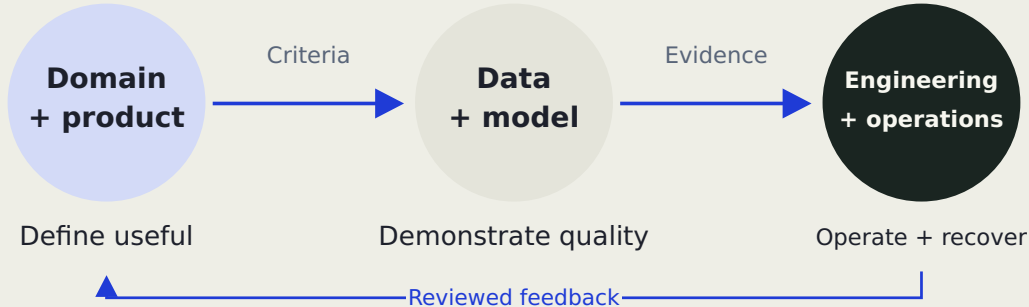
Automate the weak control, not everything.



Google automation levels; not a universal scale or a guarantee of quality.

Google Cloud levels 0–2; different from the Microsoft framework in S2 page 30.

Evidence travels with responsibility.



One person can wear several hats; the responsibilities still exist.

Safeguards belong inside the loop.



Traceable decisions support accountability; they do not prove compliance.

S1 responsible-AI concepts; S2 page 29. Engineering principles, not legal advice.

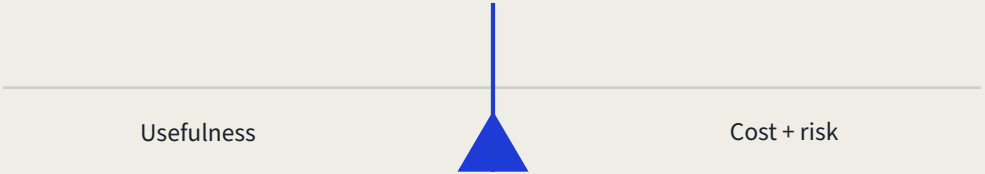
More modelling is not always more value.

A SIMPLE BASELINE

Recent generation as a starting forecast.

A LEARNED MODEL

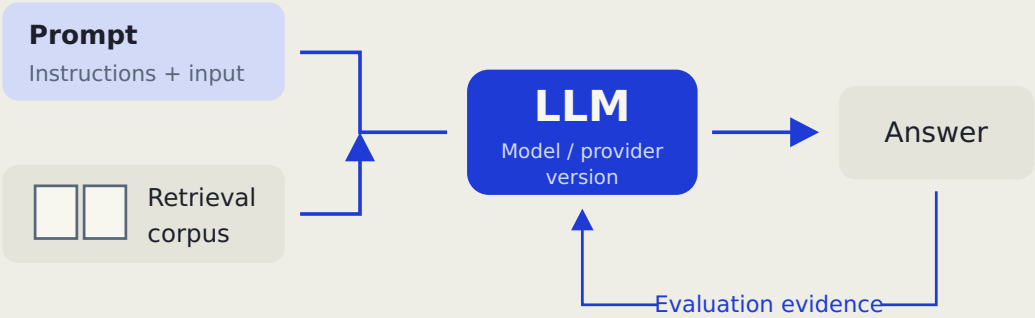
Extra predictive value has an operating cost.



Compare on the same decision and evaluation horizon.

S1 slides 51–58; S2 pages 82–84. Balance is conceptual, not measured.

The loop stays. The artefacts change.



Retrieval is one example; prompts, versions and evaluation still matter.

S2 pages 32–40. Conceptual context, not an LLM project requirement.

Reliable ML is a system property.

Controlled

Evidence before change.

Observable

Signals before decisions.

Owned

People before handoffs.

NEXT THEORY TOPIC / DATA PIPELINES

Practical slides and exercises are prepared separately.